

7.4 Analysis of missing data ↕

□ Watch: Exploring data visually (part 3) [23.09 mins]

Watch the following video, which discusses the implications of missing data and how it can be addressed. The creation of visualisations for time series data so that temporal patterns can be identified is also described.

This is useful as it enables understanding of how to deal with missing data entries and how trends, variability and seasonality in time series data can be visualised.



[Transcript](#)

<https://learn.usc.edu.au/courses/35242/files/3138650?wrap=1>

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Download the following slides for note taking: [ICT702 Exploring data visually \(part 3\)](https://learn.usc.edu.au/courses/35242/files/3168166?wrap=1) (<https://learn.usc.edu.au/courses/35242/files/3168166?wrap=1>)

Datasets often have missing values (*observations*). To know how the missing values and their associated observations should be treated, it is important to understand the reason why the values are missing.

Types of missing data

There are different types of missing data, some of which are described below.

Legitimately missing data

Missing data can occur naturally. For example, a question may not be relevant for a specific respondent; therefore, no answer may be given.

Illegitimately missing data

This occurs when respondents:

- choose not to answer a question
- drop out of a study before it is completed.

Illegitimately missing data may also occur when a sensor or other data collection device fails during a study.

When there is illegitimately missing data, it is important to consider if remedial action should be taken. There are three primary options that can be taken:

1. Discard the observations/records where missing values occur.
2. Fill in missing values with estimated values.
3. Treat missing data as a separate category (if dealing with a categorical variable).

The strategy to be used requires an understanding of why the data is missing. There are three categories:

1. *Missing completely at random* (MCAR)

The missing entry is completely unrelated to the value that is missing and unrelated to any other question (on the questionnaire).

2. *Missing at random* (MAR)

The missing entry for a variable is related to the value of another variable/s in the observation. The reason for the missing value may determine the level of criticality of the data.

3. *Missing not at random* (MNAR)

The missing entry of a variable is related to its value.

Irrespective of the reasons for the missing entries, it is important to first try to identify the actual value that is missing. This can be done by examining the data source or logically determining the likely value. If this is not possible, it is critical to accurately determine the category the missing value falls into.

□ Watch: Missing data mechanisms [6.53 mins]

This video is useful as it provides explanations of different types of missing data.

As you watch, consider:

- How might recognising the differing instances of different types of missing data influence analysis of a dataset?
- Why is evaluating the type of missing data important?



Source: © ritvikmath (2018).

A number of strategies can be used to deal with missing data. Each approach has its advantages and limitations, and the choice depends on the specific context, available data and assumptions made. Click on each heading below to learn more.

› **Dealing with MCAR values**

› **Dealing with MAR values**

› **Dealing with MNAR values**